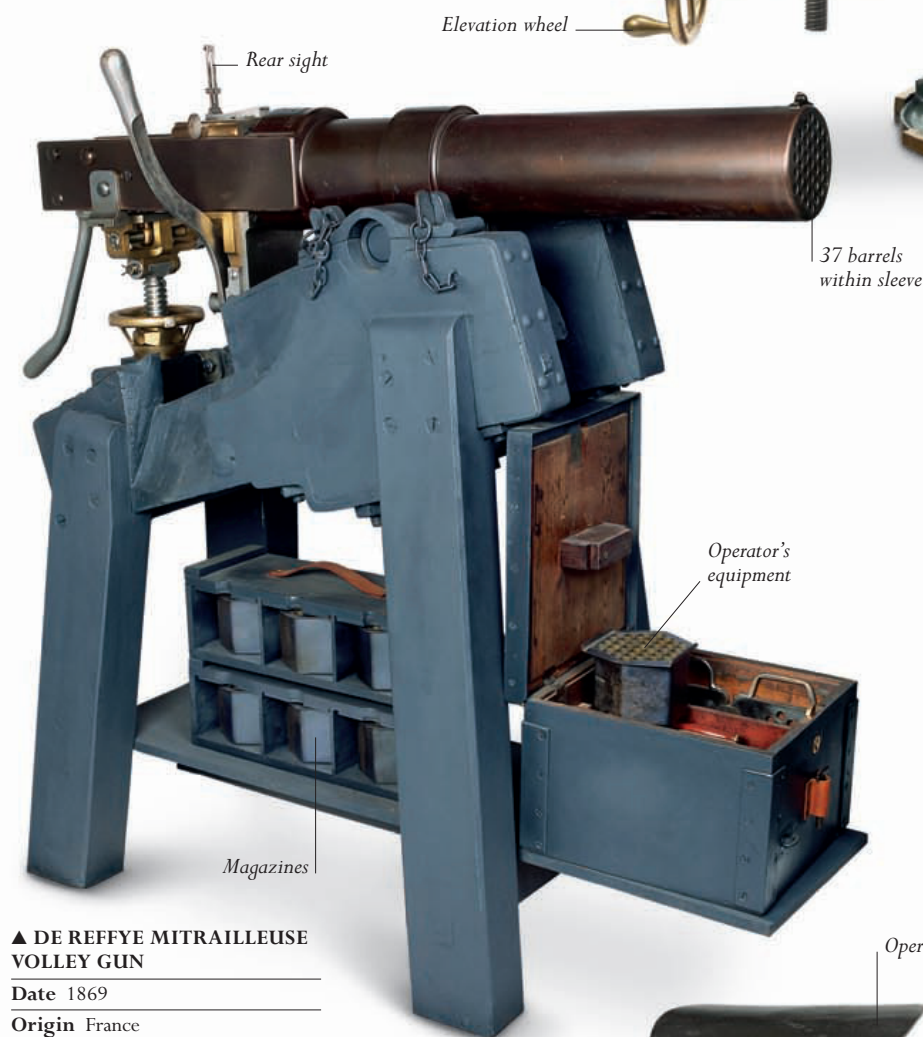


MACHINE-GUNS

The first self-powered machine-gun was developed by American Hiram Maxim in Britain in 1884. In that design, the recoil energy – the backwards force created by firing a cartridge – was used to eject the spent cartridge and then load and fire a new one from the ammunition belt. Earlier rapid-fire guns, such as the Nordenfelt and the Gardner, all relied on an operator for turning a hand-cranked lever. The Maxim was adopted in Austria and Italy in 1887, and by the British Army in 1889. Versions of the gun were manufactured in the US, Germany, Russia, and Switzerland from 1904–11. In 1896, Maxim's company was taken over by the British firm Vickers, and their improved, lighter versions of the weapon – in particular the enduring Mk 1, launched in 1912 – were to remain in service in the British Army through two world wars and beyond.



▲ DE REFFYE MITRAILLEUSE VOLLEY GUN

Date 1869

Origin France

Weight 340kg (749½lb)

Length 1.76m (5¾ft)

Calibre 13mm

Developed by Joseph Montigny of Belgium, and improved by French ordnance engineer Commandant de Reffye, this gun was used in the Franco-Prussian War (1870–71). The original was a 25-barrel weapon, while this is a 37-barrel modification. In combat, the Mitrailleuse would be mounted on a wheeled carriage.

◀ NORDENFELT GUN

Date 1873

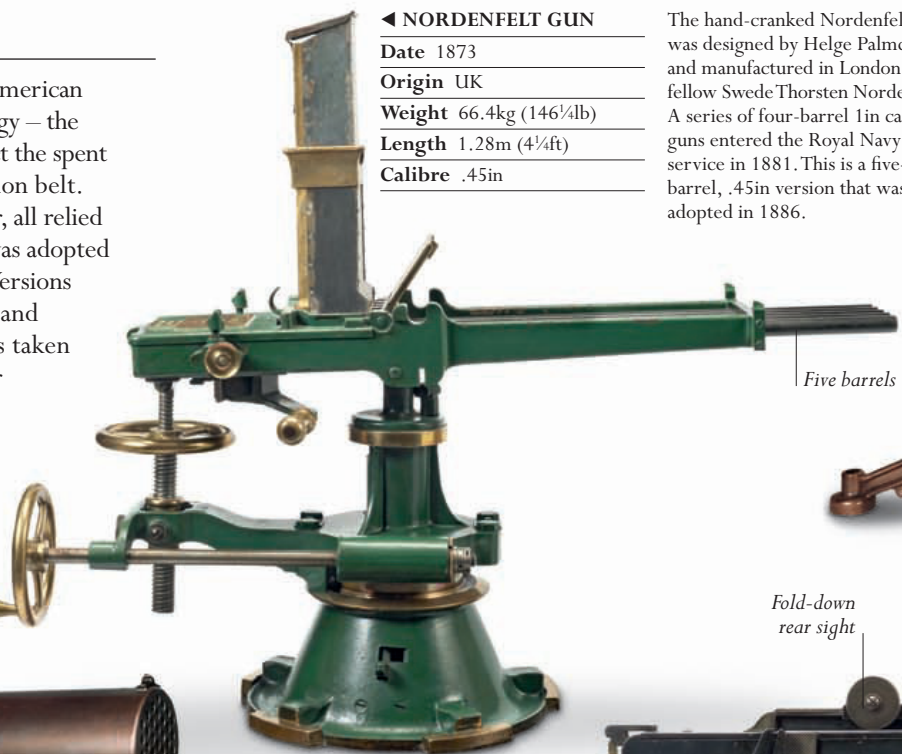
Origin UK

Weight 66.4kg (146¼lb)

Length 1.28m (4¼ft)

Calibre .45in

The hand-cranked Nordenfelt was designed by Helge Palmcrantz and manufactured in London by fellow Swede Thorsten Nordenfelt. A series of four-barrel 1in calibre guns entered the Royal Navy service in 1881. This is a five-barrel, .45in version that was adopted in 1886.



▶ MAXIM .45IN GATLING-GARDNER CALIBRE

Date 1892

Origin UK

Weight 27.12kg (59¾lb)

Length 1.12m (3¾ft)

Calibre .45in

The British Army's .45in Martini-Henry Maxim guns were converted to .303in calibre from 1897–98 onwards, but the Royal Navy remained loyal to its established .45in Gardner-Gatling calibre until the end of World War I.

